

most handset manufacturers initially refuse to implement Microsoft's operating system on their devices, Microsoft hires contract manufacturers and small designers to develop handsets that are sold directly to operators. Microsoft's operating system becomes standard, limiting device differentiation to design characteristics. Microsoft extracts mobile content revenues through licensing fees as well as by selling products and services to developers, publishers and service providers. Operators will become reliant on Microsoft, which will strongly influence the relation between offline and online content. This scenario assumes an aggressive dominance by a closed and proprietary operating system vendor as opposed to industry dominance by a consortium or a company dedicated to open standards.

### **Opportunities and Risks in the Microsoft-Dominated Scenario**

Driven by cross-platform Microsoft compatibility and synchronization, demand for handsets and connectivity will be strong in this scenario. With a technology standard on the handset, penetration may increase and the opportunity for subcomponents (graphic acceleration cards, sound cards) could emerge. Mobile device manufacturers will have to pay licenses for the Microsoft platform. The hardware trend will be toward commodity as the interface, cross-platform compatibility and standard developer interfaces pull momentum away from the hardware. Mobile operators will increasingly lose control over device specifications. They risk a "dumb pipe" future in this scenario. Operators are vulnerable to being played against each other as Microsoft extends its customer control from the offline world to the online one.

Portals, service providers, publishers and content owners would profit from a higher number of entertainment-ready handsets and a somewhat standardized online/offline platform. Premium content providers and publishers should not be negatively affected by a Microsoft-dominated scenario. Application developers may benefit from a standardized hardware and OS platform. Developers may find their margins at risk if Microsoft begins charging a license fee. Developers that fall out of favour with Microsoft risk being barred (explicitly or not) from mobile entertainment revenues. Both portals